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InsightCart

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<u>Lab Project Status</u>	
Marks:	Signature:
Comments:	Date:

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Chapter 1

Introduction

1.1 Overview

InsightCart is a machine learning-based project that focuses on analyzing customer purchase data to understand behavior and improve retail decision-making. The project uses real-world transaction data and applies different machine learning techniques to find useful patterns and insights [1].

The system processes raw data, cleans it, and creates important features such as customer activity and spending behavior. Then it applies different models to group customers, predict high-value customers, and detect unusual patterns. The main goal of this project is to turn raw data into meaningful information that can help businesses make better decisions.

1.2 Motivation

In today's retail environment, businesses collect a large amount of data from customer transactions. However, without proper analysis, this data cannot be used effectively [2]. Understanding customer behavior is very important for improving sales, customer satisfaction, and marketing strategies.

The motivation behind this project is to learn how machine learning can be used to analyze real-world data and solve practical problems [3]. By building this system, it becomes easier to understand how different models work and how they can be used to identify important customers and patterns in retail data.

1.3 Problem Definition

1.3.1 Problem Statement

Retail companies store a large amount of transaction data, but it is difficult to understand customer behavior directly from raw data. Without proper processing and analysis, it is hard to identify valuable customers, understand purchase patterns, or detect unusual activity [2].

The problem is to build a system that can clean and process retail data, create meaningful features, and apply machine learning techniques to analyze customer behavior and support better business decisions.

1.3.2 Complex Engineering Problem

Table 1.1: Summary of Attributes Related to **InsightCart**

Name of the Attribute	Explanation on How to Address
P1: Depth of knowledge required	Requires knowledge of data preprocessing, feature creation, clustering, and classification models.
P2: Range of conflicting requirements	Need to balance model accuracy and simplicity for better understanding.
P3: Depth of analysis required	Requires analyzing customer behavior and interpreting model results.
P4: Practical Constraints	Handling missing values, large data size, and noisy real-world data.

1.4 Design Goals/Objectives

The primary objective of the **InsightCart** project is to design a Machine Learning based system that can analyze customer transaction data and generate useful insights for better decision support.

- **Data Preprocessing:** Clean and prepare raw retail data by handling missing values and removing invalid records.
- **Feature Engineering:** Create meaningful features such as customer activity and spending behavior for analysis.
- **Customer Segmentation:** Group customers based on purchasing behavior using clustering techniques.
- **Predictive Modeling:** Build models to identify and predict high-value customers.
- **Anomaly Detection:** Detect unusual customer behavior or transactions.
- **Data Visualization:** Present findings using charts and summary reports for better understanding.
- **Educational Objective:** Apply Machine Learning concepts in a real-world retail data analysis problem.

1.5 Application

InsightCart can be used in different real-life and learning situations.

- **Marketing Use:** Businesses can find which customers to target for offers.
- **Sales Improvement:** Helps understand when and what products sell more.
- **Customer Management:** Helps identify loyal and inactive customers.
- **Product Planning:** Helps decide which products to promote together.
- **Risk Detection:** Helps notice unusual customer activities.
- **Learning:** Helps students practice Machine Learning using real data.

Chapter 2

Design/Development/Implementation of the Project

2.1 Introduction

Machine learning helps in finding patterns and making predictions from data. In retail systems, analyzing customer data can help businesses understand behavior and improve performance.

This project uses different machine learning techniques to process and analyze transaction data. The system takes raw data as input, prepares it for analysis, and applies different models to extract useful information. Each step is designed to gradually transform data into meaningful results.

2.2 Project Details

InsightCart can be used in different real-life and learning situations.

- **Data Preprocessing:** Clean the dataset by removing missing or invalid values.
- **Feature Engineering:** Create new features such as total spending and customer activity.
- **Exploratory Data Analysis:** Study data patterns using charts and summaries.
- **Dimensionality Reduction:** Reduce data complexity for better visualization.
- **Customer Segmentation:** Group customers based on behavior using clustering methods [4].
- **Classification Models:** Predict whether a customer is high-value or not.
- **Ensemble Learning:** Combine multiple models to improve prediction results.
- **Anomaly Detection:** Identify unusual customer behavior.

2.3 Implementation

The InsightCart project is implemented using Python in a Jupyter Notebook environment. The system uses common Machine Learning libraries such as pandas, NumPy, scikit-learn, matplotlib, seaborn, mlxtend, and XGBoost. The implementation follows a step-by-step modular structure where the dataset is loaded, cleaned, transformed, analyzed, and then used for different Machine Learning tasks such as customer segmentation, classification, ensemble learning, and anomaly detection.

The main implementation parts of the project are shown below.

Library Import

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.decomposition import PCA

from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.naive_bayes import GaussianNB

from sklearn.ensemble import (
    RandomForestClassifier, GradientBoostingClassifier,
    AdaBoostClassifier, BaggingClassifier,
    StackingClassifier, IsolationForest
)

from sklearn.neighbors import LocalOutlierFactor
from sklearn.cluster import KMeans, DBSCAN

from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, confusion_matrix, classification_report,
    roc_auc_score, roc_curve, silhouette_score
)

from mlxtend.frequent_patterns import fpgrowth, association_rules
```

Dataset Loading

```
if file_name.endswith('.xlsx'):
    df = pd.read_excel(file_name, sheet_name=None)
    df = pd.concat(df.values(), ignore_index=True)
elif file_name.endswith('.csv'):
    df = pd.read_csv(file_name, encoding='unicode_escape')
```

Data Cleaning

```
df.columns = df.columns.str.strip().str.replace(' ', '_')
df = df.dropna(subset=[customer_col])
df = df[~df[invoice_col].astype(str).str.startswith('C')]
df = df[(df[quantity_col] > 0) & (df[price_col] > 0)]
```

Outlier Removal Function

```
def remove_outliers(data, col, multiplier=1.5):
    Q1 = data[col].quantile(0.25)
    Q3 = data[col].quantile(0.75)
    IQR = Q3 - Q1
    return data[
        (data[col] >= Q1 - multiplier * IQR) &
        (data[col] <= Q3 + multiplier * IQR)
    ]
```

Feature Engineering

```
df['TotalPrice'] = df[quantity_col] * df[price_col]
df['Year'] = df[date_col].dt.year
df['Month'] = df[date_col].dt.month
df['DayOfWeek'] = df[date_col].dt.dayofweek
df['IsWeekend'] = df['DayOfWeek'].isin([5, 6]).astype(int)
```

RFM Feature Creation

```
reference_date = df[date_col].max() + pd.Timedelta(days=1)

rfm = df.groupby(customer_col).agg(
    Recency=(date_col, lambda x: (reference_date - x.max()).days),
    Frequency=(invoice_col, 'nunique'),
    Monetary=('TotalPrice', 'sum')
).reset_index()
```

Customer Segmentation Function

```
def segment_customer(row):
    r, f, m = int(row['R_Score']), int(row['F_Score']), int(row['M_Score'])

    if r >= 4 and f >= 4 and m >= 4:
        return 'Champions'
    elif r >= 3 and f >= 3 and m >= 3:
        return 'Loyal Customers'
    elif r >= 4 and f <= 2:
        return 'New Customers'
    elif r <= 2 and f >= 3 and m >= 3:
        return 'At Risk'
    elif r <= 2 and f <= 2 and m <= 2:
        return 'Lost'
    elif r <= 2 and f >= 4:
        return 'Cannot Lose Them'
    else:
        return 'Others'
```

PCA

```
scaler = StandardScaler()
rfm_scaled = scaler.fit_transform(rfm[['Recency', 'Frequency', 'Monetary']])

pca = PCA()
pca_result = pca.fit_transform(rfm_scaled)
```

Association Rule Mining

```
basket = df.groupby([invoice_col, desc_col])[quantity_col].sum().unstack().fillna(0)
basket = basket.applymap(lambda x: 1 if x >= 1 else 0)

frequent_itemsets = fpgrowth(basket, min_support=0.01, use_colnames=True)
rules = association_rules(frequent_itemsets, metric='lift', min_threshold=1.0)
```

K-Means Clustering

```
clustering_scaled = scaler.fit_transform(
    rfm[['Recency', 'Frequency', 'Monetary']]
)

kmeans = KMeans(n_clusters=optimal_k, random_state=42, n_init=10)
rfm['KMeans_Cluster'] = kmeans.fit_predict(clustering_scaled)
```

Classification Model

```
X = rfm[['Recency', 'Frequency', 'Monetary']]
y = rfm['HighValue']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

model = RandomForestClassifier(random_state=42, n_estimators=100)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Model Evaluation

```
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

cm = confusion_matrix(y_test, y_pred)
```

Ensemble Learning

```
stacking_model = StackingClassifier(
    estimators=[
        ('rf', RandomForestClassifier(random_state=42)),
        ('svm', SVC(probability=True)),
        ('knn', KNeighborsClassifier())
    ],
    final_estimator=LogisticRegression()
)

stacking_model.fit(X_train, y_train)
```

Anomaly Detection

```
iso_forest = IsolationForest(contamination=0.05, random_state=42)
rfm['Anomaly_IF'] = iso_forest.fit_predict(anomaly_scaled)

lof = LocalOutlierFactor(n_neighbors=20, contamination=0.05)
rfm['Anomaly_LOF'] = lof.fit_predict(anomaly_scaled)

rfm['Anomaly_Combined'] = (
    (rfm['Anomaly_IF'] == -1) & (rfm['Anomaly_LOF'] == -1)
).astype(int)
```

Final Result Summary

```
print("InsightCart Project Summary")
print("Total Customers:", rfm.shape[0])
print("Total Transactions:", df[invoice_col].nunique())
print("Total Sales:", df['TotalPrice'].sum())
print("Customer Segments:", rfm['Cluster'].nunique())
print("High Value Customers:", rfm['HighValue'].sum())
print("Anomalies Detected:", len(anomalies))
```

The above code snippets represent the core implementation of the InsightCart project, highlighting the main steps from data preprocessing to advanced analytics. Key techniques such as RFM-based customer segmentation, clustering, association rule mining, classification modeling, and anomaly detection are applied to extract meaningful insights. These components collectively demonstrate how data-driven methods can be used to understand customer behavior, improve decision-making, and enhance overall business performance.

2.4 Algorithms

In the InsightCart project, several Machine Learning algorithms are used to analyze customer data and perform different tasks such as grouping customers, predicting high-value customers, and detecting unusual behavior.

2.4.1 K-Means Clustering

K-Means is used to group customers based on Recency, Frequency, and Monetary values.

Algorithm 1: K-Means Clustering for Customer Segmentation

Input: Customer feature set (Recency, Frequency, Monetary)

Output: Cluster labels for each customer

- 1 Initialize K cluster centers randomly;
 - 2 Repeat until convergence::
 - 3 Assign each customer to the nearest cluster center;
 - 4 Update cluster centers by calculating mean of assigned points;
 - 5 Return final clusters;
-

2.4.2 Random Forest Classification

Random Forest is used to predict whether a customer is high-value or not.

Algorithm 2: Random Forest Classification

Input: Training data (X, y)

Output: Predicted class labels

- 1 Select multiple random samples from training data;
 - 2 Build decision trees for each sample;
 - 3 Combine predictions from all trees;
 - 4 Return final prediction based on majority voting;
-

2.4.3 FP-Growth Algorithm

FP-Growth is used to find frequent itemsets and product associations.

Algorithm 3: FP-Growth for Association Rule Mining

Input: Transaction dataset

Output: Frequent itemsets and rules

- 1 Scan dataset and find frequent items;
 - 2 Construct FP-Tree;
 - 3 Generate conditional patterns;
 - 4 Extract frequent itemsets;
 - 5 Generate association rules;
-

2.4.4 Isolation Forest

Isolation Forest is used to detect anomalies in customer behavior.

Algorithm 4: Isolation Forest for Anomaly Detection

Input: Customer feature dataset

Output: Anomaly labels

- 1 Randomly select features and split values;
 - 2 Build isolation trees;
 - 3 Measure path length for each data point;
 - 4 If path length is short, mark as anomaly;
 - 5 Return anomaly results;
-

2.4.5 Model Evaluation

Different evaluation metrics are used to measure model performance.

Algorithm 5: Model Evaluation

Input: True labels, Predicted labels

Output: Performance metrics

- 1 Calculate Accuracy;
 - 2 Calculate Precision;
 - 3 Calculate Recall;
 - 4 Calculate F1-score;
 - 5 Generate Confusion Matrix;
-

Chapter 3

Performance Evaluation

3.1 Simulation Environment/ Simulation Procedure

The InsightCart project is implemented in a Jupyter Notebook environment using Python programming language. The system uses common machine learning libraries such as pandas, NumPy, scikit-learn, matplotlib, and seaborn. This environment allows easy step-by-step execution and clear visualization of results.

At the beginning, the dataset is loaded and checked for structure, missing values, and data types. The dataset used in this project is the Online Retail II dataset collected from Kaggle [5].. After cleaning, new features are created to represent customer behavior.

The system then applies different machine learning models for clustering, classification, and anomaly detection. Each model is trained and tested using the same dataset to ensure fair comparison. The dataset is split into training and testing sets where necessary.

Finally, the results are evaluated using different performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix. Visualization is also used to better understand the results and patterns.

3.2 Results Analysis/Testing

The results of the project show that machine learning models can effectively analyze customer behavior and provide useful insights. Different models perform differently depending on the task.

Clustering methods successfully group customers into different segments based on their behavior. These groups help in understanding which customers are more active or valuable.

Classification models are able to predict high-value customers with good performance. Some models perform better than others, and comparing them helps in selecting the best model.

Anomaly detection methods identify unusual customers who behave differently from others. These customers may represent special cases or require further analysis.

Overall, the system shows that applying machine learning techniques on retail data can provide meaningful and useful results.

3.2.1 Result_portion

The results of the **InsightCart** are presented through selected screenshots and visualizations generated from the analysis. These outputs highlight key insights such as customer behavior, sales trends, segmentation, model performance, and anomaly detection.

upload the online_retail_II.xlsx or .csv file

[Choose Files](#) online_retail_II.xlsx

online_retail_II.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 45628287 bytes, last modified: 4/27/2026 - 100% done

Saving online_retail_II.xlsx to online_retail_II.xlsx

Figure 3.1: online_retail_II.xlsx dataset uploaded successfully

***	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country	
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-12-01 07:45:00	6.95	13085.0	United Kingdom	
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom	
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom	
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-12-01 07:45:00	2.10	13085.0	United Kingdom	
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-12-01 07:45:00	1.25	13085.0	United Kingdom	
5	489434	22064	PINK DOUGHNUT TRINKET POT	24	2009-12-01 07:45:00	1.65	13085.0	United Kingdom	
6	489434	21871	SAVE THE PLANET MUG	24	2009-12-01 07:45:00	1.25	13085.0	United Kingdom	
7	489434	21523	FANCY FONT HOME SWEET HOME DOORMAT	10	2009-12-01 07:45:00	5.95	13085.0	United Kingdom	
8	489435	22350	CAT BOWL	12	2009-12-01 07:46:00	2.55	13085.0	United Kingdom	
9	489435	22349	DOG BOWL , CHASING BALL DESIGN	12	2009-12-01 07:46:00	3.75	13085.0	United Kingdom	

Figure 3.2: Preview of the dataset showing first 10 rows

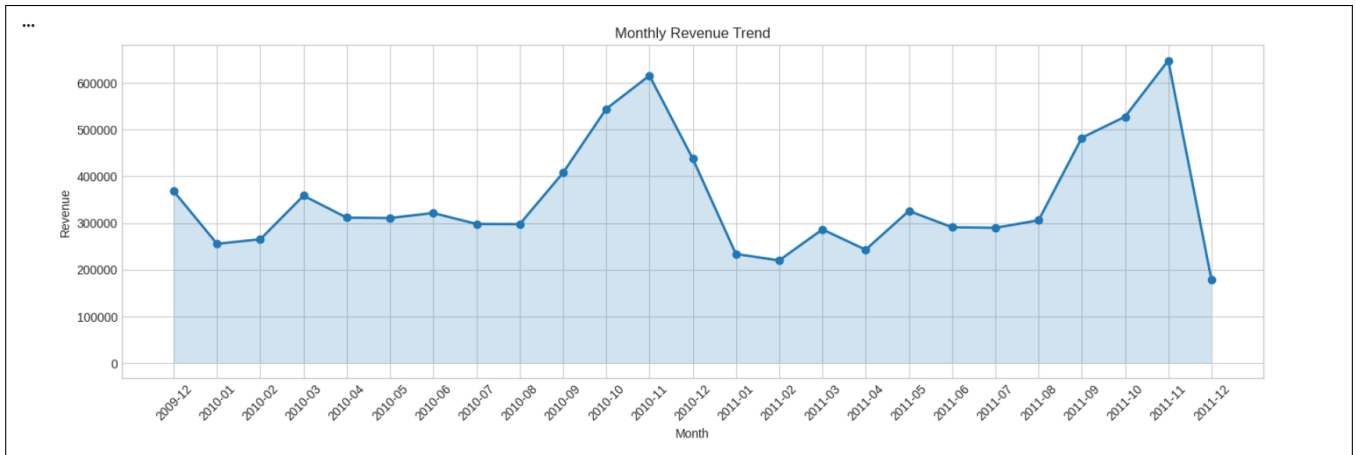


Figure 3.3: Monthly revenue trend showing sales fluctuations and seasonal patterns over time

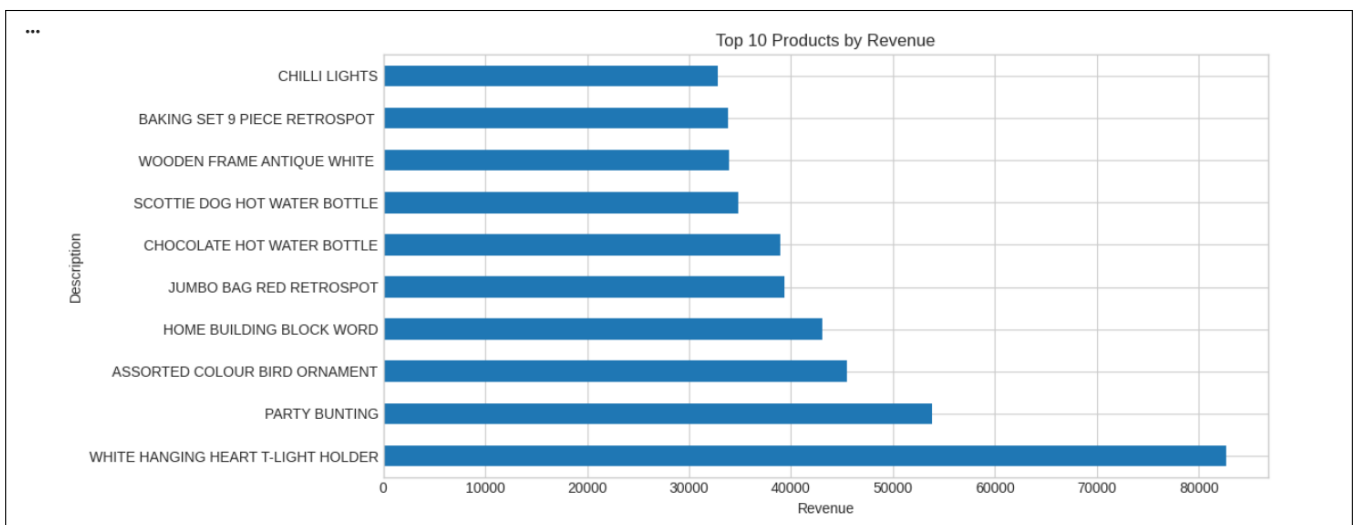


Figure 3.4: Top 10 products ranked by total revenue contribution

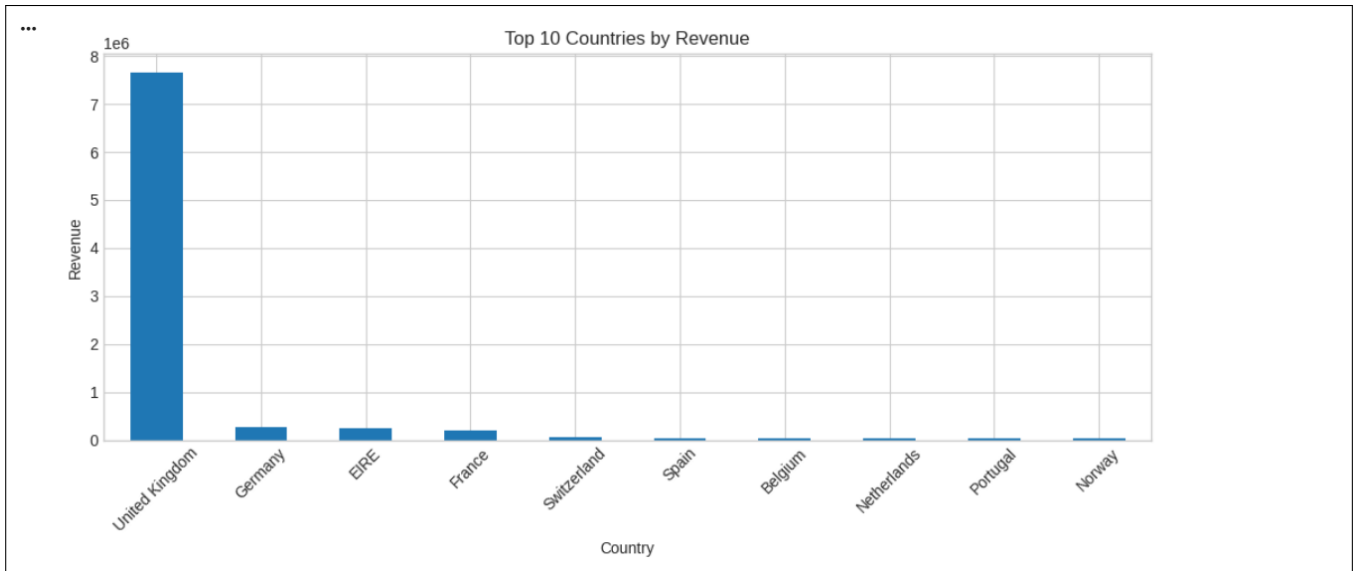


Figure 3.5: Top 10 countries contributing to total revenue, highlighting dominant markets

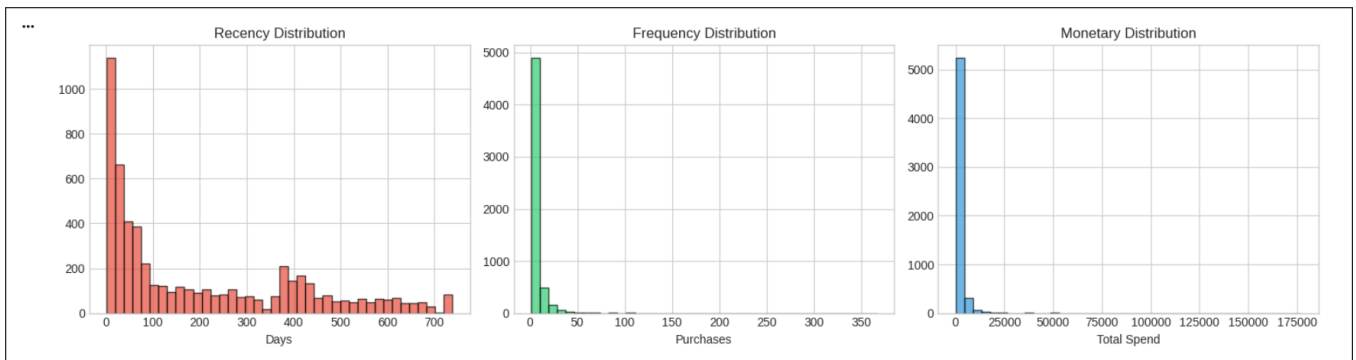


Figure 3.6: Distribution of Recency, Frequency, and Monetary values showing customer purchasing behavior patterns

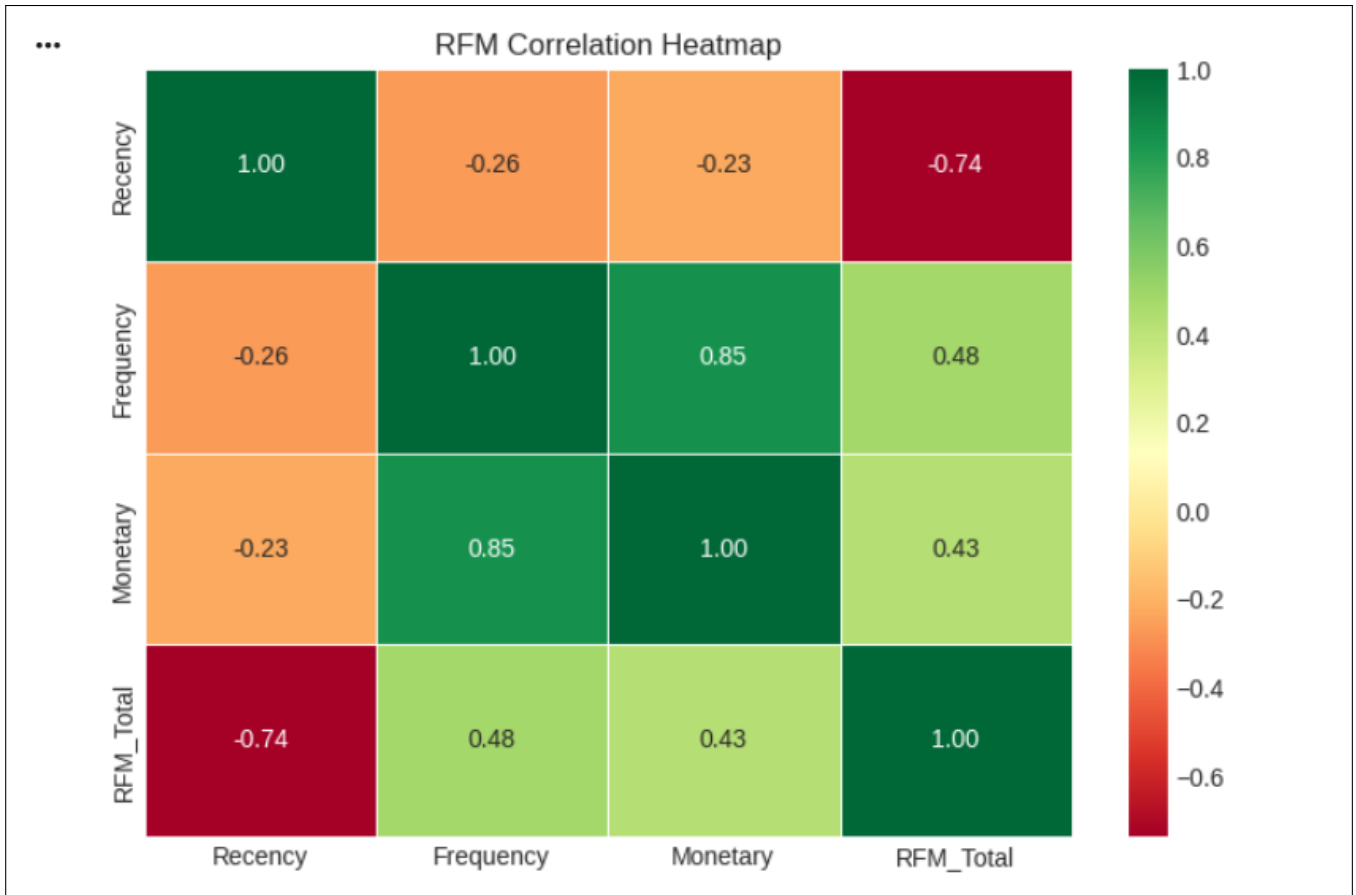


Figure 3.7: Correlation heatmap showing relationships among Recency, Frequency, Monetary, and overall RFM score

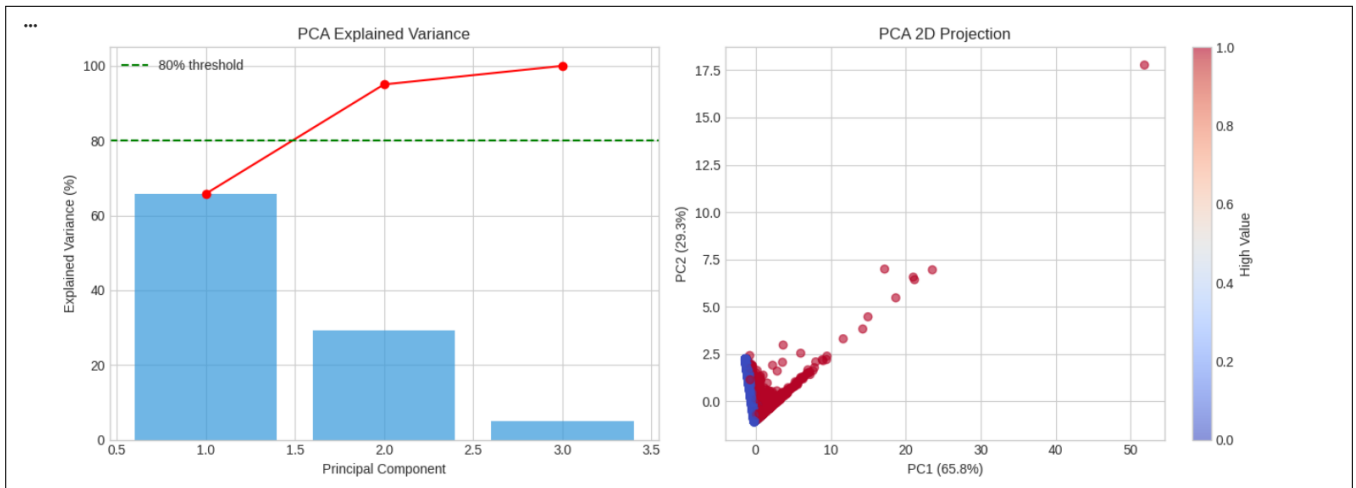


Figure 3.8: PCA results showing explained variance and 2D projection of customers based on RFM features

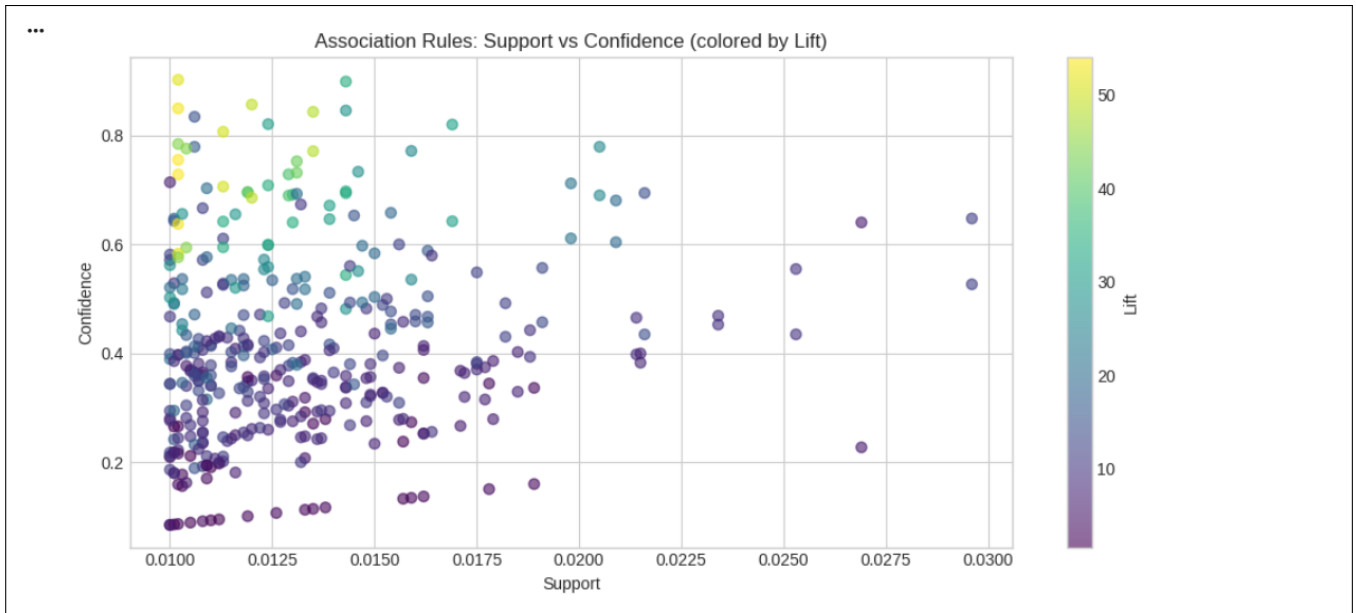


Figure 3.9: Association rules visualization showing relationship between support and confidence, with lift indicating rule strength

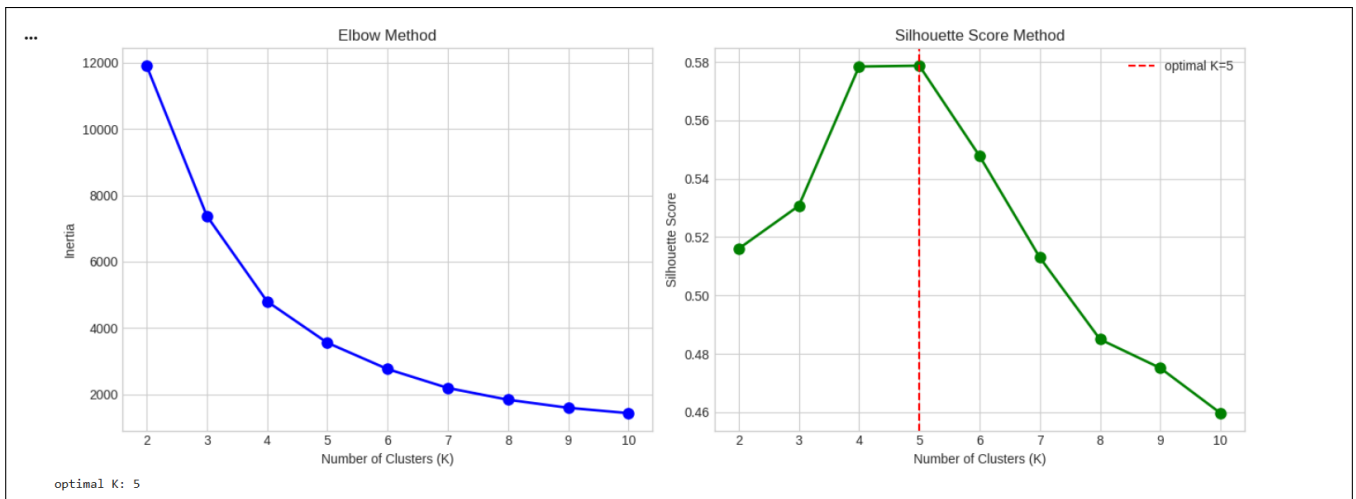


Figure 3.10: Elbow and Silhouette methods used to determine the optimal number of clusters (K=5)

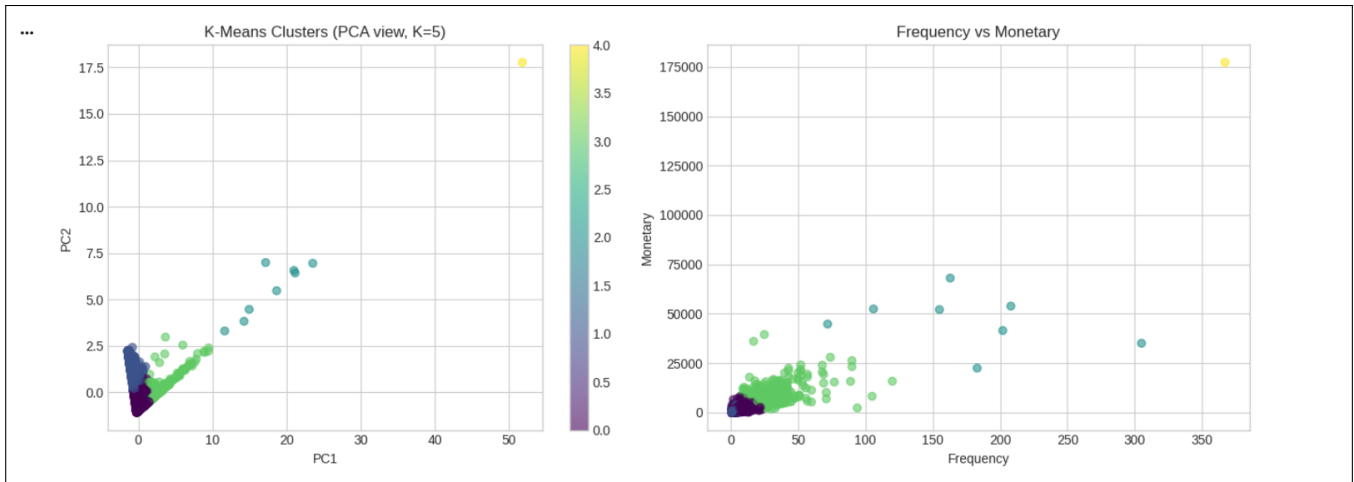


Figure 3.11: K-Means clustering visualization showing customer segments based on RFM features (K=5)

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
1	Decision Tree	1.0000	1.0000	1.0000	1.0000	1.0000
2	Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000
6	Gradient Boosting	1.0000	1.0000	1.0000	1.0000	1.0000
0	Logistic Regression	0.9806	0.9946	0.9665	0.9803	0.9985
5	SVM	0.9797	0.9945	0.9647	0.9794	0.9991
3	KNN	0.9762	0.9770	0.9753	0.9762	0.9949
4	Naive Bayes	0.9286	0.9841	0.8713	0.9242	0.9893

Figure 3.12: Comparison of classification models based on accuracy, precision, recall, F1-score, and ROC-AUC metrics

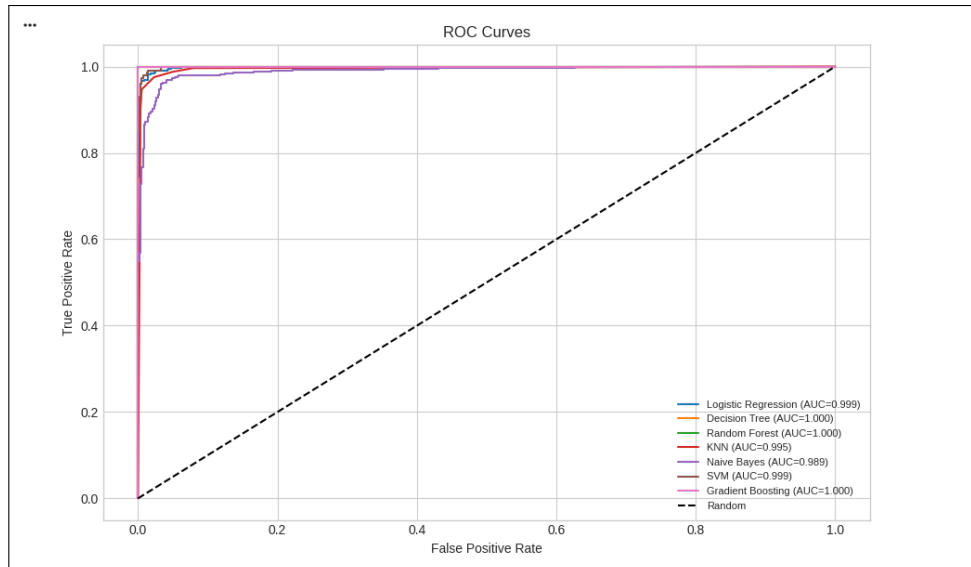


Figure 3.13: ROC curves comparing classification model performance, showing high predictive accuracy across models

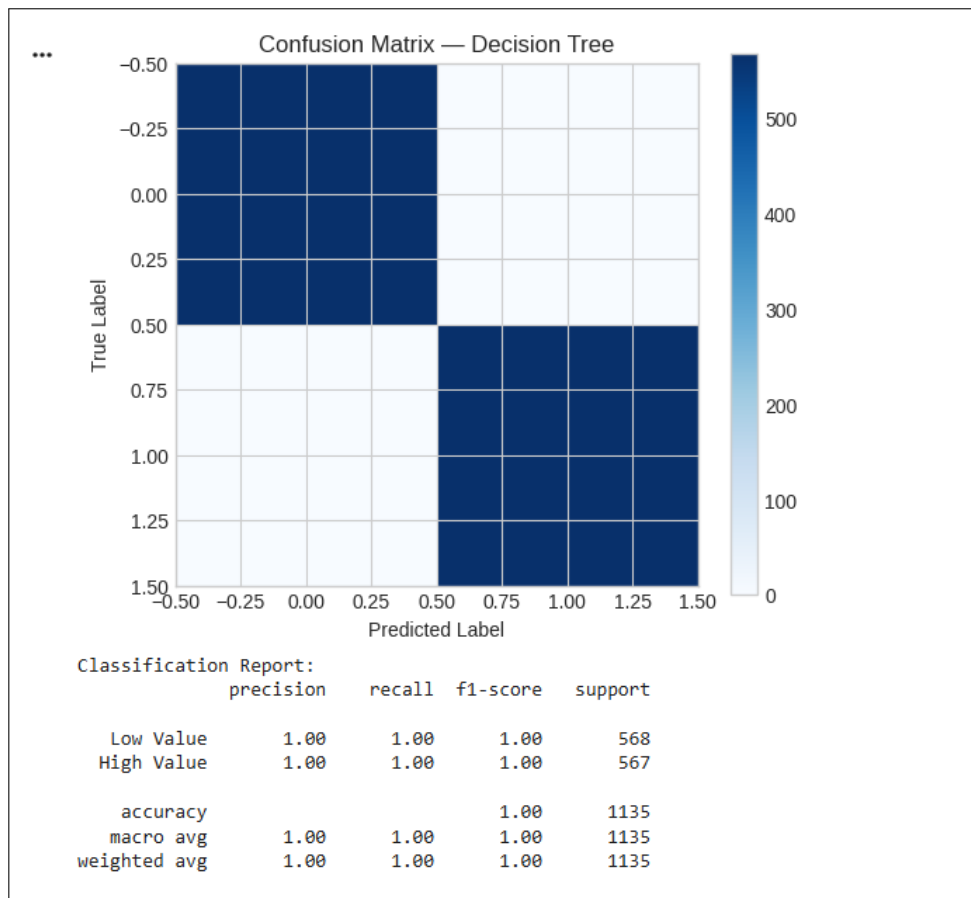


Figure 3.14: Confusion matrix and classification report for the best model (Decision Tree), showing perfect prediction performance

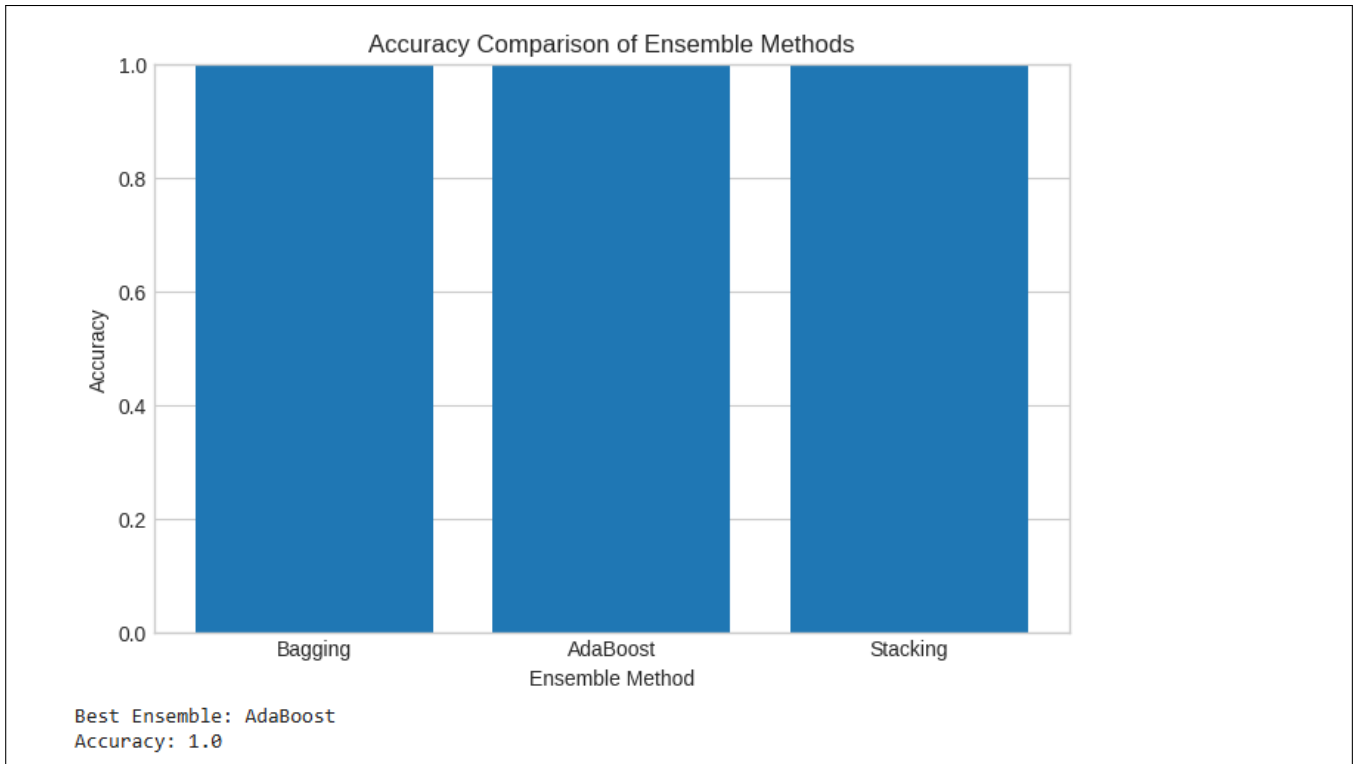


Figure 3.15: Comparison of ensemble methods showing AdaBoost as the best-performing model with highest accuracy

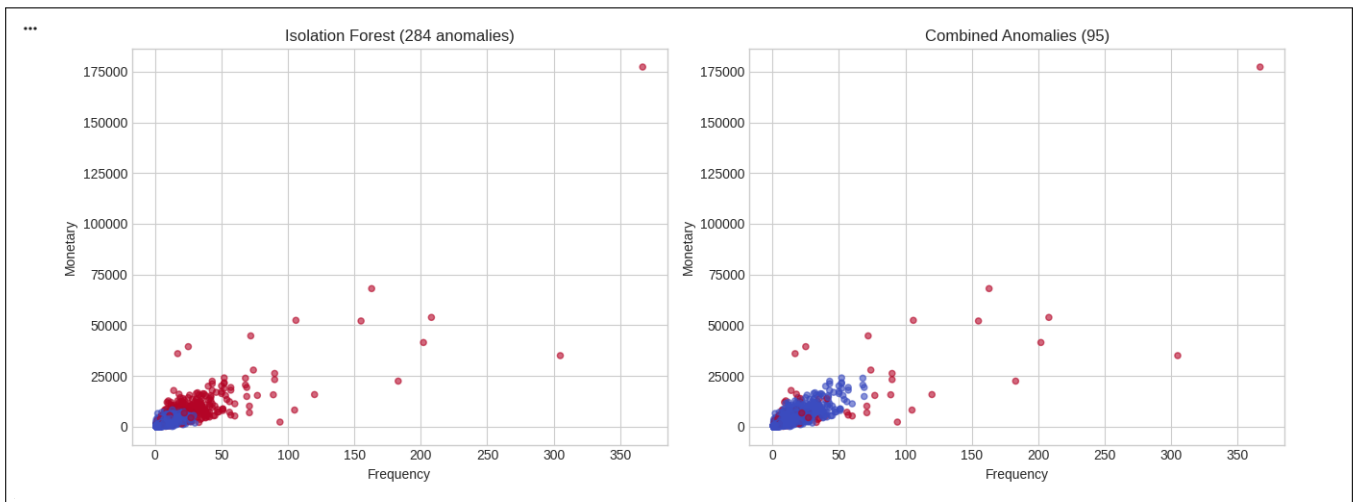


Figure 3.16: Anomaly detection results highlighting unusual customer behavior using Isolation Forest and combined methods

```

*** =====
                SUMMARY - InsightCart
=====

Dataset Overview
-----
Original rows:      1067371
Clean rows:         686764
Unique Customers:   5675
Total Revenue:      8825609.21

Association Rules (FP-Growth)
-----
Frequent Itemsets:  692
Total Rules:         384
Strong Rules:        278

Customer Segmentation (K-Means)
-----
Optimal K:           5
Silhouette Score:    0.5787
  Champions: 1240 (21.9%)
  Lost: 1204 (21.2%)
  Loyal Customers: 1088 (19.2%)
  Others: 1079 (19.0%)
  At Risk: 593 (10.4%)
  New Customers: 445 (7.8%)
  Cannot Lose Them: 26 (0.5%)

Best Classification Model
-----
Model:      Decision Tree
Accuracy:    1.0
F1-Score:    1.0
ROC-AUC:     1.0

Anomaly Detection
-----
Isolation Forest:  284
LOF:                284
Combined:           95 (1.67%)

=====
                COMPLETED
=====

```

Figure 3.17: Final project summary highlighting key insights from data analysis, segmentation, modeling, and anomaly detection

```

...  rfm results saved to rfm_analysis_results.csv

```

Figure 3.18: RFM analysis results successfully saved as a CSV file for further use

3.3 Results Overall Discussion

The overall results show that the InsightCart project successfully analyzes customer transaction data and provides clear insights about customer behavior. Each module plays an important role in building the final system.

Clustering helps in grouping customers, which makes it easier to understand different types of customers. Classification models help in predicting important customers, which can be useful for business strategies.

The combination of multiple models improves the overall performance of the system. By comparing different models, it becomes easier to choose the most suitable one for a specific task.

Visualization also plays an important role by making the results easier to understand. Graphs and charts clearly show patterns and differences in the data.

Overall, the project demonstrates how machine learning can be used to transform raw data into useful information for better decision-making.

3.3.1 Complex Engineering Problem Discussion

The InsightCart project includes different Machine Learning tasks, so it is not a simple problem. It requires working with real-world data, which often contains missing values and errors. Cleaning and preparing this data is an important part of the project.

The system also uses different models like clustering, classification, and anomaly detection. Choosing the right model and understanding the results can be challenging. Some models are easy to understand but less accurate, while others are more complex.

Another challenge is analyzing the results properly. The outputs must be explained in a simple way so that they can be useful for decision-making.

There are also practical issues such as handling large data and processing time. These factors make the project more complex.

Overall, the project combines data processing, model building, and analysis, which makes it a complete and meaningful engineering problem.

Chapter 4

Conclusion

4.1 Discussion

The InsightCart project shows how machine learning techniques can be used to analyze customer data and extract useful insights. The system processes raw data step by step and applies different models to understand customer behavior.

By using clustering, the project groups customers into different categories. Classification models help in predicting high-value customers, which is important for business planning. Anomaly detection helps in identifying unusual patterns in customer behavior.

Each part of the system contributes to building a complete solution. The project not only improves understanding of machine learning techniques but also shows how they can be applied in real-world situations.

4.2 Limitations

Although the InsightCart project provides useful results, there are some limitations that affect its performance and practical use. These limitations are mainly related to the dataset, model behavior, and system design.

- The project is based on a specific dataset, so results may vary with different data.
- Feature selection may affect model performance.
- Large datasets may increase processing time.
- Some unusual patterns may not always represent real problems.

Overall, these limitations show that the system works well for learning and analysis, but further improvements are needed for real-world deployment.

4.3 Scope of Future Work

The InsightCart project can be improved in several ways to make it more effective and practical. Future improvements can help increase accuracy, usability, and real-world application.

- Use larger and more diverse datasets for better generalization.
- Apply advanced machine learning or deep learning models.
- Improve feature selection techniques.
- Build a real-time prediction system.
- Add a user interface for easier interaction.
- Improve model accuracy through hyperparameter tuning.

These future improvements will make the system more powerful and suitable for real-world retail applications.

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